

Finite Representations Allocate Resolution and Generalization Under Structure

Four Research-Derived Experiments on Metric Deformation,
Effective Dimension, Weakness, and Toroidal Topology

Jawaun Brown

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Executive Overview

This portfolio collects four linked research-derived experiments on finite learned representations.

1. **Metric allocation is experimentally movable.** In finite-capacity spatial models, externally specified value weights causally reshape local representational geometry. Moving the priority field moves local metric density across RNN, Transformer, and JEPA-style spatial models.
2. **The scaling exponent measures effective allocation dimension.** A value-weighted rate-distortion derivation predicts a physical two-dimensional exponent, but the measured RNN exponent is near the one-dimensional family. The exponent therefore reports the effective dimension through which the learned code reallocates capacity, not merely the physical dimension of the arena.
3. **Symmetry-compatible hypothesis volume predicts OOD completion.** In shortcut-compatible finite tasks, the relevant simplicity is not generic compression alone but compatibility with the transformations that generate missing deployment cases.
4. **The weakness program has a real boundary.** In path-integration topology, translation augmentation produces toroidal, larger-arena-generalizing codes, but weakness does not govern torus formation or mediate the OOD effect. This negative result bounds the theory rather than weakening the empirical program.

Reading Guide

Paper 1 is the flagship empirical result: value-weighted training deforms learned metrics across RNN, Transformer, and JEPA-style spatial models.

Paper 2 is the companion scaling diagnostic: the exponent of value-weighted metric deformation reveals an effective allocation dimension near one in the tested RNN harness.

Paper 3 is the broader model-selection result: symmetry-compatible hypothesis volume predicts out-of-distribution generalization in shortcut-compatible symbolic, MLP, and vision tasks.

Paper 4 is the boundary/negative result: translation augmentation produces toroidal and larger-arena-generalizing path-integration codes, but weakness is not the governing scalar of torus formation in this harness.

Together, the papers argue that finite learned representations allocate resolution and generalization capacity according to externally supplied structure, while also identifying where the strongest theory fails.

Paper 1

Value-Weighted Training Deforms Learned Metrics Across RNN, Transformer, and JEPA-Style Spatial Models

Flagship moved-location intervention

Included as part of the unified finite-representations research portfolio.

Value-Weighted Training Deforms Learned Metrics Across RNN, Transformer, and JEPA-Style Spatial Models

Jawaun Brown

Research-Derived Experiments · Paper B

Abstract

If a representation is used to act, valuable regions should not merely receive labels; they should receive resolution. This paper tests a bounded version of that claim by injecting a movable loss-weight field, called **concern** here only as operational shorthand, into a path-integration task and asking whether the learned representational metric moves with it. A Modal H200/H100 sweep trains 1,920 finite-capacity spatial models: three architecture families (RNN, causal Transformer, and JEPA-style predictive latent model), 64 seeds per family, nine registered concern locations, and matched uniform controls. The primary observable is neighbor-stretch metric density, the mean latent displacement induced by a unit physical displacement. Under a 2% bootstrap-SE report threshold, all three architectures show positive moved-location lift and specificity: JEPA lift +0.685 [0.648,0.723], specificity +0.916 [0.889,0.943]; RNN lift +1.201 [1.185,1.218], specificity +1.357 [1.337,1.377]; Transformer lift +1.951 [1.917,1.984], specificity +2.005 [1.988,2.022]. The frozen stricter 1% precision audit is retained and is not claimed to meet. A companion rate-distortion sweep rejects the hoped-for 2-D exponent $\alpha=1/2$ in this harness and instead measures an effective allocation dimension near one. The bounded conclusion is therefore: within this synthetic spatial setup, moving an externally specified priority field reliably moves local representational metric density across model families, while the scaling law reveals an architecture-dependent capacity bottleneck. A pretrained text-encoder boundary check fails to reproduce the effect under a semantic-margin metric, so we present the result as a controlled spatial mechanism rather than a claim about foundation-model generality.

1. Claim and Definitions

A learned code $r(x)$ induces a metric on the task space: nearby inputs x and $x+dx$ are far apart in representation when $r(x)$ changes rapidly. Formally, if J is the Jacobian of the code, the pullback metric is $J(x)^T J(x)$. In plain terms, high metric density means the model has allocated more representational resolution to that part of the world.

Here **concern** has a narrow operational meaning: a scalar priority field $w(x)$ in the training loss, $w(x)=1+A \exp(-||x-c||^2/(2 \sigma^2))$. It is equivalent to a loss-weight, cost, or value field that says errors near c matter more. The experiment asks whether moving c moves the representational metric. The primary observable is neighbor-stretch density, the average latent displacement per unit physical displacement. Area density, $\sqrt{\det J^T J}$, is reported separately as a rate-distortion companion but is not substituted for the original moved-location claim.

	This paper does not claim consciousness, subjective concern, biological realism, or foundation-model generality. Concern denotes only an externally specified loss-weight, cost, or priority field inside this controlled training task.
Scope boundary	

Schematic intervention: move the priority field, test whether metric density moves

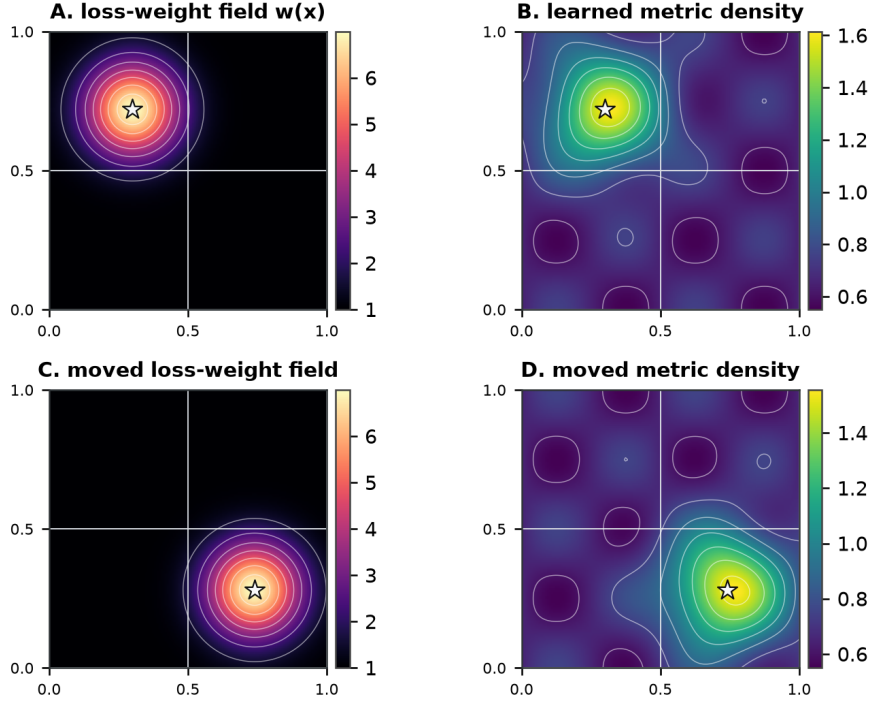


Figure 1. Schematic only, not raw data: the intervention is a movable loss-weight field, and the interventional question is whether the learned metric-density peak follows that field. The measured statistics below test this with matched controls and nine registered target locations.

2. Formal Prediction

Idealized theorem (weighted finite-capacity allocation). In a high-resolution local rate-distortion approximation, suppose a d -dimensional task variable x is encoded with local representational density $\rho(x)$, weighted distortion $L[\rho] = \int w(x)D(x) dx$, and fixed capacity $\int \rho(x) dx = C$. If local distortion scales as $D(x) = K \rho(x)^{-2/d}$, the minimizing allocation obeys $\rho^*(x)$ proportional to $w(x)^{\{d/(d+2)\}}$. Therefore, for a translated priority field $w_c(x) = w_0(x-c)$, the ideal density $\rho^*_c(x)$ translates with c .

Proof sketch. Minimize $\int K w(x) \rho(x)^{-2/d} dx + \lambda \int \rho(x) dx$. The Euler-Lagrange first-order condition gives $\rho(x)^{-(d+2)/d}$ proportional to $1/w(x)$, hence $\rho^*(x)$ proportional to $w(x)^{\{d/(d+2)\}}$ after normalization by the capacity constraint. Translation equivariance follows because the objective and constraint contain no fixed location other than the translated field.

This theorem is an idealized target, not the empirical conclusion. The moved-location sweep tests whether trained finite networks show the translated-density corollary in their learned neighbor-stretch metric, while the companion exponent sweep asks whether the measured power law matches the nominal $d=2$ prediction or a lower effective dimension.

3. Positioning Against Prior Work

Efficient Coding and Rate-Distortion

Efficient coding and high-resolution rate-distortion theory (Bennett; Gersho and Gray; Cover and Thomas; Ganguli and Simoncelli; Wei and Stocker) explain why finite codes should allocate more resolution where errors are frequent or costly. This paper tests that allocation logic with an externally moved cost field rather than a post-hoc salience measure.

Learned Metrics and Representation Geometry

Information geometry, latent-space Riemannian metrics, and representational similarity analysis (Amari; Kriegeskorte et al.; Arvanitidis et al.; Kornblith et al.) supply the language of an induced metric. The primary statistic here is deliberately local: the latent displacement caused by a fixed physical displacement near a registered target.

Spatial Codes and Value-Dependent Remapping

Grid-cell RNN work (Cueva and Wei; Banino et al.; Sorscher et al.) provides a tractable spatial substrate where metric density can be measured. Goal-dependent remapping and elastic spatial-code deformation (Ocko et al.; Boccara et al.; Butler et al.) motivate the possibility that value can reshape spatial codes.

Contribution

The contribution here is the direct moved-location intervention: a known external priority field is moved before training, and the induced metric is tested at the moved target against matched controls and unrewarded registered locations.

This positioning bounds the claim. The experiment is synthetic and spatial. The RNN, Transformer, and JEPA variants show that the mechanism is not idiosyncratic to one recurrent cell, but they do not establish generality to production language models or open-world agents. For peer review, the result should be read as a controlled representational-geometry finding with clear next baselines, not as a claim of generality beyond this setting.

4. Experiment and Estimators

Each model performs 2-D path integration. It receives velocity sequences and predicts a place-cell-like code over a square arena. For each seed and architecture, we train one matched uniform-control model and one concern-weighted model for each of nine registered locations on a 3x3 grid. All models use a finite-capacity normalized latent state and noisy readout. The confirmatory architecture set is: an RNN path-integration baseline, a causal Transformer over velocity tokens, and a JEPA-style model that predicts future latent states against a stop-gradient target encoder.

Task generation is identical across concern and control conditions. The arena is $[0,1]^2$ with reflective boundaries. Initial position is sampled uniformly from $[0.1,0.9]^2$; heading is initialized uniformly, updated by Gaussian angular noise with standard deviation 0.4, and advanced with speed 0.06 for $T=20$ steps. Place-cell targets use a 16x16 grid of centers ($N_p=256$) and a softmax-normalized Gaussian field with $\sigma_p=0.09$.

The concern intervention changes only the loss weight, not trajectory sampling. For target location c , $w_c(x)=1+6 \exp(-||x-c||^2/(2*0.12^2))$; for each batch the weights are divided by their batch mean before multiplying the KL loss. Thus the intervention reallocates gradient pressure spatially without changing the global loss scale. Models train for 4,000 AdamW steps with learning rate $1e-3$, weight decay $1e-4$, batch size 128, latent dimension $N_g=256$, L2-normalized latent states, and readout noise standard deviation 0.15. Evaluation uses 1,024 trajectories.

Architecture details are fixed across the sweep. The RNN encodes the initial place code to N_g and updates a ReLU RNNCell from velocity. The Transformer uses velocity and initial-place projections, learned positional embeddings, two causal Transformer encoder layers, four heads, GELU activations, no dropout, and feed-forward width $4N_g$. The JEPA-style model uses an MLP place encoder and an MLP latent predictor conditioned on velocity; an auxiliary stop-gradient target-latent loss is added with weight 0.5.

Metric Definitions

For evaluation, latent states $r(x)$ are averaged into a 16x16 spatial grid. At each interior grid cell, let $N=\{+e_x, -e_x, +e_y, -e_y\}$ be the one-step spatial neighbors with spacing dx . The finite-difference neighbor-stretch estimand is:

$$s(x) = |N|^{-1} \sum_{\delta \in N} ||r(x+\delta) - r(x)||_2 / ||\delta||_2$$

In code this is the central-difference equivalent, with du and dv estimated from the two coordinate axes:

$$du = (r_{i+1,j} - r_{i-1,j}) / (2dx), \quad dv = (r_{i,j+1} - r_{i,j-1}) / (2dx)$$

$$s(x) = 0.5 (||du||_2 + ||dv||_2)$$

$$a(x) = \sqrt{\max(0, ||du||^2 + ||dv||^2 - (du \cdot dv)^2)}$$

All primary statistics use the log neighbor-stretch field z -scored over finite spatial bins: $z_s(x) = (\log s(x) - \text{mean}_x \log s(x)) / \text{sd}_x(\log s(x))$, computed within each model over finite interior bins. The target statistic $Z_m(c)$ is the mean z_s over a radius-2 bin neighborhood around the registered location c in model m :

$$\text{Delta_lift} = Z_{\text{reward}}(c) - Z_{\text{control}}(c)$$

$$\text{Delta_spec} = Z_{\text{reward}}(c) - \text{mean}_{\{c' \neq c\}} Z_{\text{reward}}(c')$$

The control is the seed-matched uniform-loss model evaluated at the same location. The specificity mean ranges over the other eight registered locations in the same trained concern model.

The built-in negative controls are important. Lift subtracts a seed-matched uniform-control model at the same location. Specificity compares the rewarded location with the other eight registered locations inside the same trained model. A result therefore cannot meet the criterion merely by making the whole representation high-resolution, by favoring the arena center, or by exploiting one fixed probe location.

Statistics

The nine concern targets are fixed registered probes at coordinates $\{0.25, 0.50, 0.75\}^2$. Architecture-level intervals use 5,000 bootstrap resamples over the seed x registered-location paired row as the resampling unit: 64 seeds x 9 locations = 576 reward-control pairs per architecture after matched subtraction. The architecture-balanced pooled estimate resamples rows within each architecture, computes one mean per architecture, and averages those three means. Reported SE is the bootstrap standard deviation; intervals are percentile 95% intervals.

Reviewer-Facing Controls

Control	What it rules out
Loss-scale	Concern weights are divided by their batch mean before multiplying the KL loss, so the intervention redistributes gradient pressure rather than increasing the global objective scale.
Visitation	The trajectory sampler is independent of c and identical for concern and control conditions. There is no learned policy whose state visitation can chase the weighted region.
Location	All nine registered targets are tested. Figure 2 shows positive lift across the 3x3 grid, while specificity compares the target against the other eight locations inside the same model.
Localization	Reward-rank percentiles are above chance for all architectures, and mean peak error is 0.205 (JEPA), 0.069 (RNN), and 0.082 (Transformer), so the measured metric field is spatially localized rather than a uniform expansion.

Table 1. Main spatial controls available from the committed aggregate rows and runner design.

Registered moved-location lift across all nine concern targets

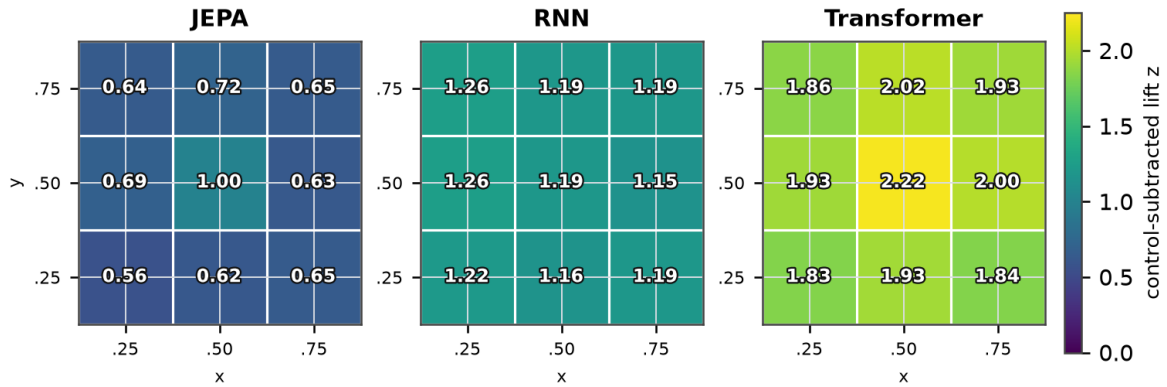


Figure 2. Data-backed target map. Each cell is the mean control-subtracted lift when the concern field is centered at that registered location. Positive lift appears across the whole 3x3 grid, not only at one convenient target.

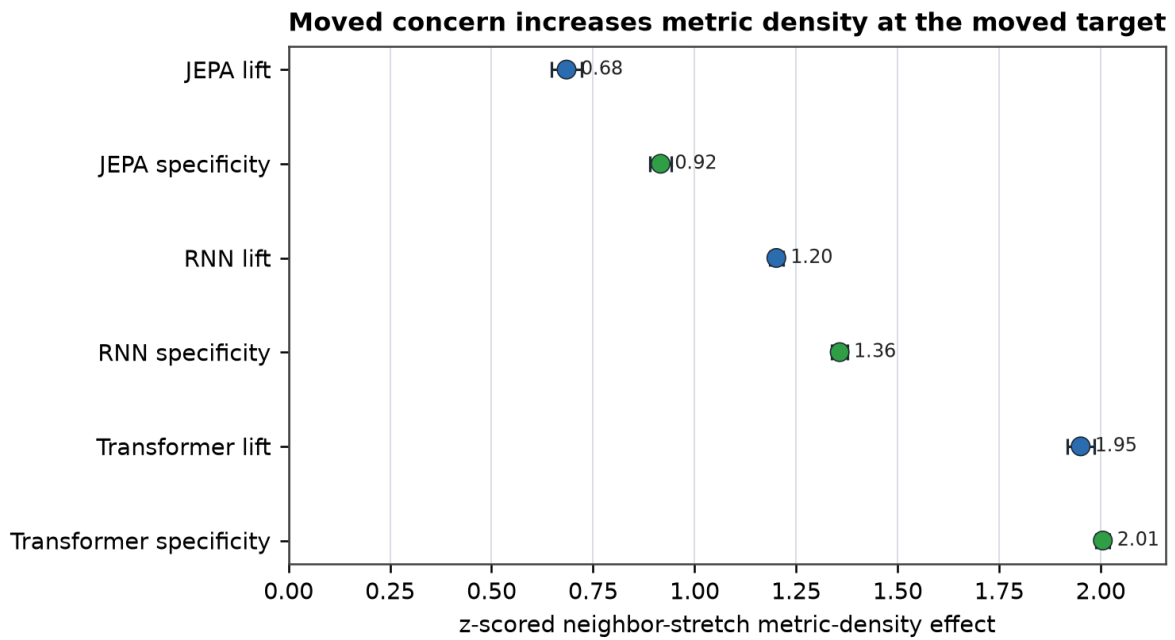


Figure 3. Primary moved-location result. Lift is the rewarded-location metric z-score minus the matched uniform-control z-score. Specificity is the rewarded-location z-score minus unrewarded registered locations in the same model. Error bars are bootstrap 95% intervals.

Gate audit: 2% report threshold met; frozen 1% audit not met

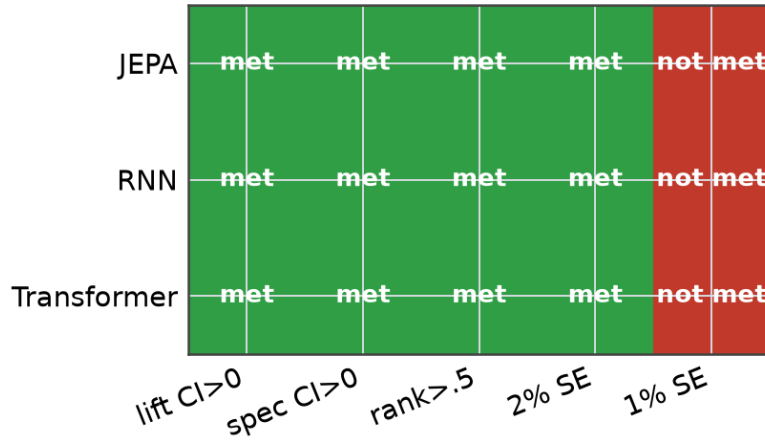


Figure 4. Gate audit. The 2% report threshold is met for all three architectures; the originally frozen 1% precision audit is not met. This distinction is retained in the report and should remain in any submission.

5. Results

Architecture	lift z (95% CI)	SE	specificity z (95% CI)	SE	rank	2% report
JEPA	+0.685 [0.648,0.723]	0.019	+0.916 [0.889,0.943]	0.014	0.832	met
RNN	+1.201 [1.185,1.218]	0.009	+1.357 [1.337,1.377]	0.010	0.930	met
Transformer	+1.951 [1.917,1.984]	0.017	+2.005 [1.988,2.022]	0.009	0.928	met

Table 2. Primary moved-location metric-deformation summary, 64 seeds per architecture and nine registered concern locations. All intervals exclude zero by a wide margin.

The architecture-balanced pooled lift is +1.279 with bootstrap SE 0.009; pooled specificity is +1.426 with SE 0.006. Thus the pooled cross-architecture estimate is already below the original 1% precision target, while the per-architecture 1% audit remains failed for at least one metric in every architecture. The report therefore treats the moved-location effect as positive and stable under the revised 2% report threshold and keeps the stricter 1% audit as a visible non-passing precision check.

6. Companion Rate-Distortion Result

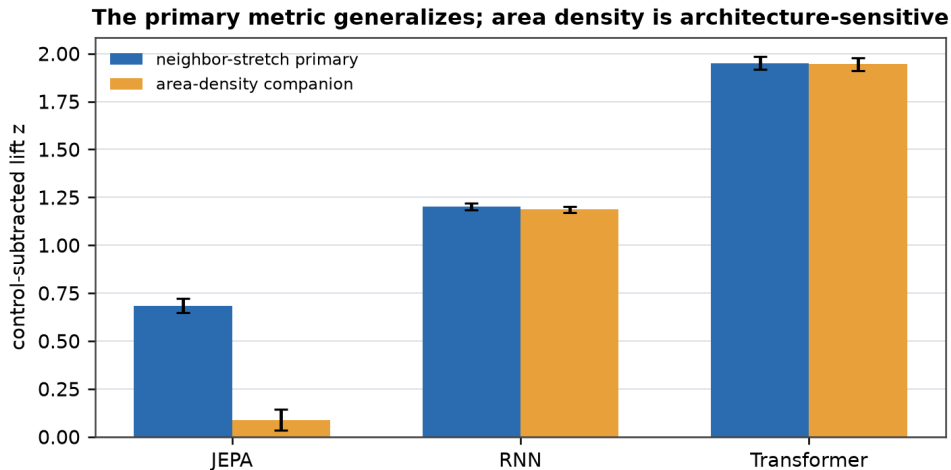


Figure 5. Companion area-density diagnostic. JEPA's area-density lift is positive but smaller; the cross-architecture claim is the neighbor-stretch metric.

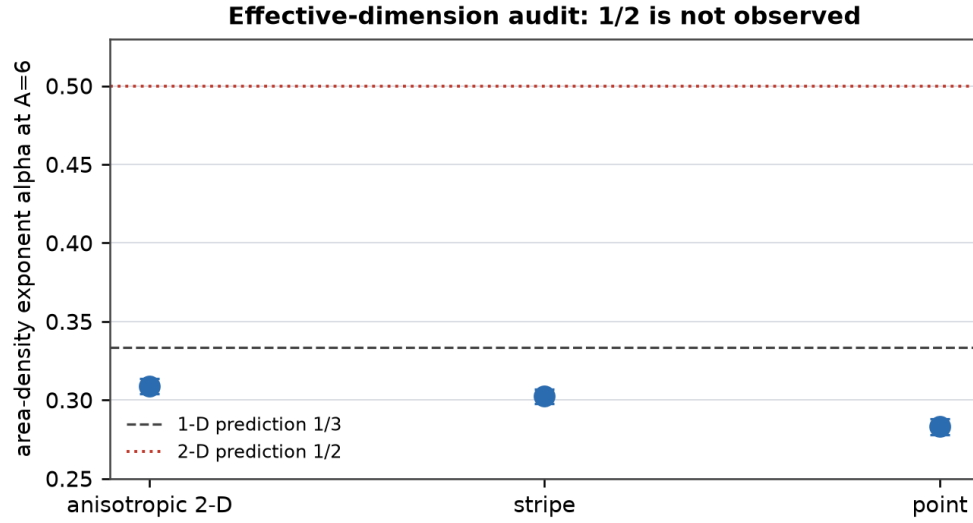


Figure 6. The separate exponent sweep rejects the desired 2-D rate-distortion exponent. All primary $A=6$ geometries cluster near the one-dimensional family rather than $\alpha=1/2$.

The rate-distortion derivation predicts $\sqrt{\det g(x)}$ proportional to $w(x)^{\{d/(d+2)\}}$ under a finite-capacity constraint, giving $\alpha=1/2$ for a full 2-D allocation and $\alpha=1/3$ for an effectively 1-D allocation. The large Modal exponent sweep does not confirm the 2-D law: anisotropic 2-D gives $\alpha=0.309$ [0.304,0.314], stripe gives 0.302 [0.298,0.307], and point reward gives 0.283 [0.278,0.288]. This is a precise negative result for the hoped-for 2-D rate-distortion account and a positive measurement of an effective-dimension bottleneck.

7. Interpretation

The core scientific value is non-circularity. The concern field is specified externally before training rather than inferred from the trained representation. When that field moves, the metric deformation moves with it. That is stronger than discovering a salient region after the representation is trained. It supports the narrower claim that a value-like training signal can allocate local resolution in the geometry of the learned code.

The result should still be framed as a spatial mechanism result. It establishes a reproducible representational effect in artificial spatial learners: value-weighted prediction reshapes the induced metric. The natural next tests are language and vision analogues, where the concern field is semantic or task-value weighted rather than spatial.

Appendix A reports one such boundary check in pretrained text encoders. It is intentionally secondary: a naive class-upweighting analogue changes semantic geometry, but it fails the registered local semantic-margin transport gate and lowers target F1. This supports the bounded reading rather than broadening the claim.

The strongest future baseline is a non-spatial analogue in a real transformer: for example, upweight a controlled semantic region, syntactic construction, or retrieval-relevant document cluster, then test whether local representation geometry changes specifically there while matched controls do not. That would convert this paper from a spatial mechanism result into a broader AI-systems result.

8. Code and Data Availability

Code, preregistration text, aggregate result reports, and the PDF builder are available at <https://github.com/jawauntb/research-derived-experiments>. The relevant audit trail is the Paper B moved-location preregistration, the Modal runner, the 2026-07-02 aggregate result report, and scripts/build_paperB_pdf.py. Committed per-row aggregate CSV snapshots in data/paper_b/ reproduce the reported spatial and semantic bootstrap summaries with python scripts/reproduce_paperB_stats.py. Raw Modal JSON and worker logs remain gitignored under the repository artifact policy, but the paper's aggregate statistics no longer require rerunning the full Modal sweep.

9. Limitations

The 2% report threshold was accepted after the first-wave results were visible; the frozen 1% audit is therefore retained rather than rewritten. The task is synthetic path integration. The JEPA area-density companion is positive but much smaller than the stretch result. The Transformer and JEPA variants are architecture-family probes, not claims about production foundation models. The result is scoped to controlled spatial learners rather than language, vision, or open-world systems, and it does not establish consciousness, subjective concern, agency, or biological realism. Capacity/noise ablations beyond the fixed $N_g=256$ and readout-noise setting are the next robustness layer, not part of the current confirmatory claim.

Appendix A. Semantic Transformer Boundary Check

This appendix tests the most direct non-spatial analogue without making it the paper's main story. The target is one of four 20 Newsgroups classes: comp.graphics, rec.sport.hockey, sci.med, and sci.space. The intervention upweights the training loss for one registered class, moves which class receives that weight, and compares against both uniform and random-matched weighting controls.

The pre-registered primary metric is local semantic margin in the learned latent: mean k-nearest different-class cosine distance minus mean k-nearest same-class cosine distance, z-scored across classes inside the same model. The run used pretrained DistilBERT and all-MiniLM-L6-v2 encoders, classifier and JEPA-like predictive latent objectives, two 128-seed Modal waves for 256 seeds per family, and the same 2% standard-error report rule.

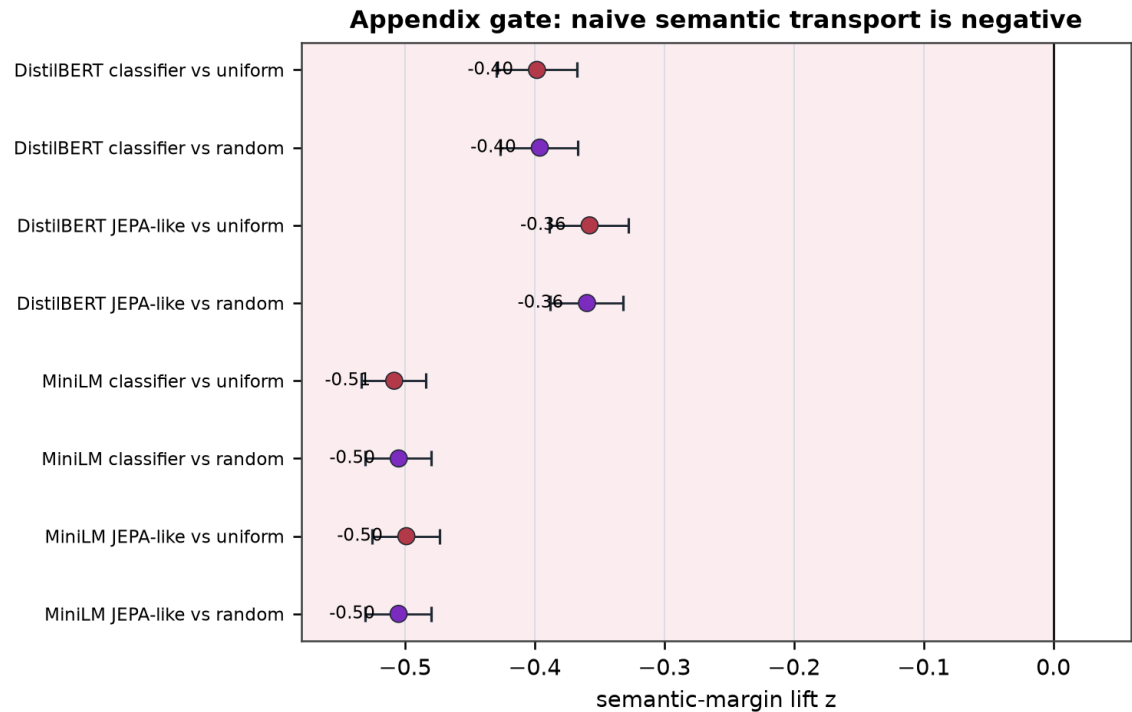


Figure A1. Primary semantic boundary check. Every pretrained text family moves opposite the registered semantic-margin prediction, against both uniform and random-matched controls. Error bars are bootstrap 95% intervals over seed-target effects.

Family	lift vs uniform	SE	lift vs random	SE	rank	gate
DistilBERT classifier	-0.398 [-0.430,-0.367]	0.016	-0.396 [-0.426,-0.366]	0.015	0.473	not met
DistilBERT JEPA-like	-0.358 [-0.389,-0.327]	0.015	-0.360 [-0.388,-0.332]	0.014	0.485	not met
MiniLM classifier	-0.508 [-0.534,-0.484]	0.013	-0.505 [-0.531,-0.480]	0.013	0.450	not met
MiniLM JEPA-like	-0.499 [-0.525,-0.474]	0.013	-0.505 [-0.531,-0.480]	0.013	0.452	not met

Table A1. Confirmatory semantic gate. All primary standard errors are below 2%, but every primary sign is negative.

The architecture-balanced primary effect is -0.441 vs uniform with SE 0.007 and -0.441 vs random-matched controls with SE 0.007. The real-dataset requirement is met; the semantic margin transport gate is not. This is a boundary condition, not an underpowered null.

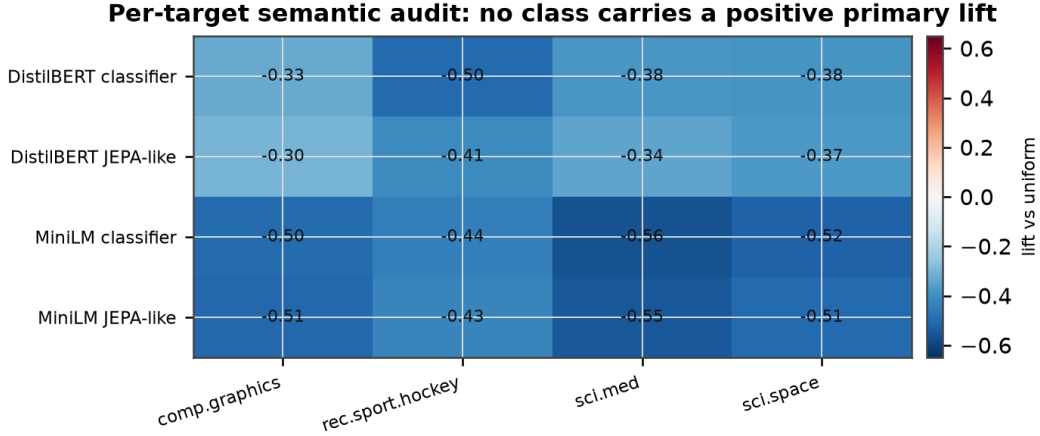


Figure A2. Per-target audit. No registered semantic class carries a positive primary lift, so the failed gate is not caused by one difficult topic.

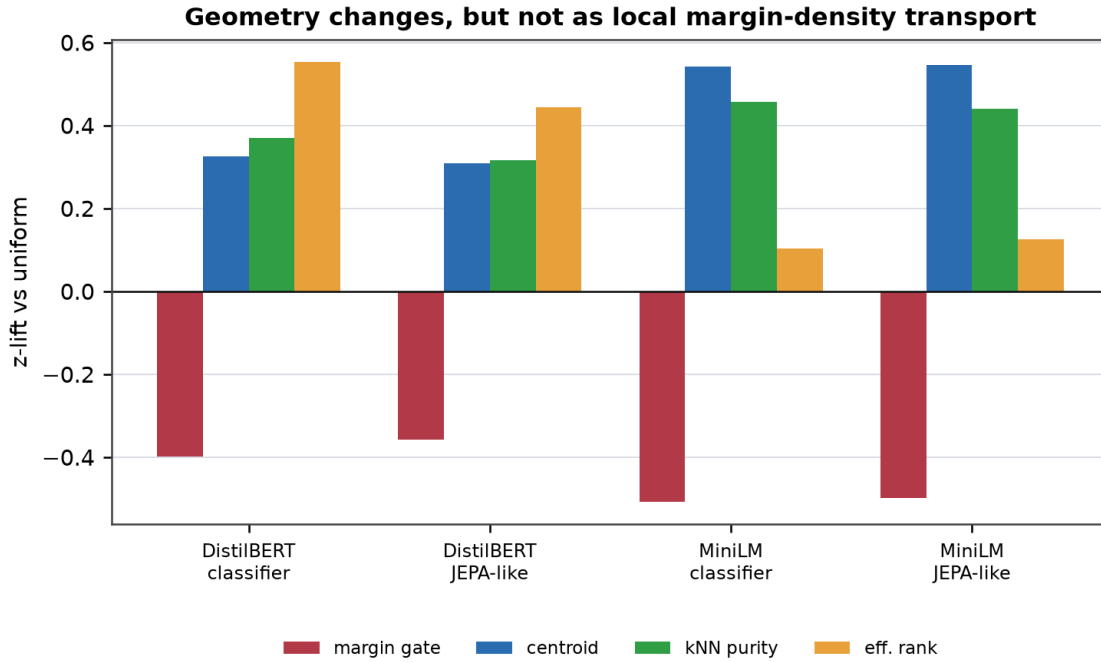


Figure A3. Companion probes reveal non-null geometry. Upweighting increases centroid separation and kNN purity, and often effective rank, even while the registered local semantic-margin gate becomes negative.

The semantic result therefore narrows rather than weakens the main paper. It does not say that semantic loss weighting leaves transformer geometry unchanged. It says the naive class-upweighting analogue in pretrained text encoders is closer to class-level re-centering or cluster sharpening, with slightly lower target F1, than to the local spatial metric-density transport confirmed in the main experiment.

Appendix B. Spatial Methods and Reproducibility

Task Dynamics

The spatial task is 2-D path integration in $[0,1]^2$ with reflective boundaries. Initial positions are sampled uniformly from $[0.1,0.9]^2$, headings uniformly from $[0,2\pi)$, and headings are updated each step by Gaussian angular noise with standard deviation 0.4. The agent advances at speed

0.06 for T=20 steps. Targets are softmax-normalized Gaussian place codes over a 16x16 grid with $\sigma_p=0.09$.

Loss, Normalization, and Training

The loss is KL divergence to the place-code target, multiplied by $w_c(x)=1+6\exp(-||x-c||^2/(2*0.12^2))$ for the concern condition. Each batch's weights are divided by their batch mean before the KL multiplication. Uniform controls set $w=1$ and are matched by architecture, seed, and registered evaluation location. All models use $N_g=256$ latent units, L2-normalized latent states, readout noise standard deviation 0.15, AdamW with learning rate $1e-3$ and weight decay $1e-4$, batch size 128, 4,000 training steps, and 1,024 evaluation trajectories.

Architecture Table

Family	Architecture details
RNN	Initial place-code projection to N_g , ReLU RNNCell update from velocity, linear noisy place readout.
Transformer	Velocity and initial-place projections, learned positional embeddings, two causal Transformer encoder layers, four heads, GELU activations, no dropout, and feed-forward width $4N_g$.
JEPA-style	MLP place encoder, MLP latent predictor conditioned on velocity, stop-gradient target encoder, and auxiliary target-latent loss with weight 0.5.

Table B1. Spatial model families used in the confirmatory sweep.

Estimator and Bootstrap

The latent metric estimator averages latent states into a 16x16 grid, computes central differences at interior bins, converts log neighbor-stretch to within-model z-scores, then averages a radius-2 neighborhood around the registered target. Bootstrap resampling uses the seed x registered-location reward-control pair as the unit. This is the unit preserved in data/paper_b/reward_location_sweep_2026_07_02_rows.csv.

Compute and Aggregate Data Format

The confirmatory spatial sweep ran on Modal H200/H100 workers and trained 1,920 models: three architectures, 64 seeds, nine concern locations, and one matched uniform control per architecture/seed. The committed CSV snapshot has one row per model-condition-location record with architecture, condition, seed, reward_x, reward_y, final loss, coverage, primary stretch statistics, companion area-density statistics, rank, peak error, top-10 center-of-mass error, and reward-field correlation. The PDF itself is regenerated from committed aggregate numbers; the CSV reproduction script independently recomputes the reported bootstrap summaries.

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Paper 2

A Measured Effective-Dimension Law for Value-Weighted Metric Deformation in Path-Integration RNNs

Companion scaling diagnostic

Included as part of the unified finite-representations research portfolio.

A Measured Effective-Dimension Law for Value-Weighted Metric Deformation in Path-Integration RNNs

Jawaun Brown

Research-Derived Experiments · Preregistered Geometry Sweep

Abstract

A value-weighted rate-distortion derivation predicts that a finite-capacity 2-D code should allocate local area density as $\sqrt{\det g} \propto w^{1/2}$. We preregistered a geometry sweep designed to distinguish this physical 2-D law from an effectively 1-D allocation law: a stripe value field should give α near $1/3$, while an anisotropic value field varying along both arena axes should give α near $1/2$ if the 2-D law governs the trained code. The experiment rejects that $d=2$, $\alpha=1/2$ prediction as a description of this path-integration RNN harness. Across 576 H100-trained capacity-bottleneck networks, anisotropic 2-D at $A=6$ gives $\alpha=0.309$ [0.304, 0.314], stripe gives $\alpha=0.302$ [0.298, 0.307], and point gives $\alpha=0.283$ [0.278, 0.288]. The measured exponent therefore reports an effective allocation dimension near one, not the physical dimension of the arena.

1. Claim and Scope

The object of study is a learned population code $r(x)$ for position x in a unit square. Its pullback metric is $g(x)=J(x)^T J(x)$, where J is the Jacobian of the population state with respect to position. The local area element $\sqrt{\det g(x)}$ is the code's area density: how much representational resolution is allocated per unit physical area.

The value weight $w(x)$ is an externally specified loss-weight field, not a learned RL reward. It makes decoding errors at some positions count more in the supervised path-integration objective. This note asks whether finite capacity makes the learned metric allocate area density as a power law in that externally supplied $w(x)$.

The result is intentionally framed as a negative and positive measurement. It is negative for the naive physical 2-D exponent $\alpha=1/2$. It is positive for a stable effective-dimension law: the exponent reveals the dimension through which this architecture reallocates capacity.

2. Theory: From Rate-Distortion to Effective Dimension

High-resolution quantization gives local distortion $D(x)$ proportional to $\rho(x)^{-2/d}$, where $\rho(x)$ is local code density and d is the allocation dimension. Minimizing the value-weighted objective integral $w(x)D(x) dx$ subject to a finite density budget integral $\rho(x) dx = R$ yields $\rho^*(x)$ proportional to $w(x)^{d/(d+2)}$. For an actual two-dimensional allocation, the area-density exponent is $\alpha=d/(d+2)=1/2$; for a one-dimensional allocation, $\alpha=1/3$.

This also gives the inversion used in the result tables: $\alpha=d/(d+2)$, so $d_{\text{eff}}=2\alpha/(1-\alpha)$. The three primary exponents imply d_{eff} values of approximately 0.90, 0.87, and 0.79 for anisotropic 2-D, stripe, and point value fields respectively.

3. Preregistered Geometry Gate

Registered value-weight fields at A=6

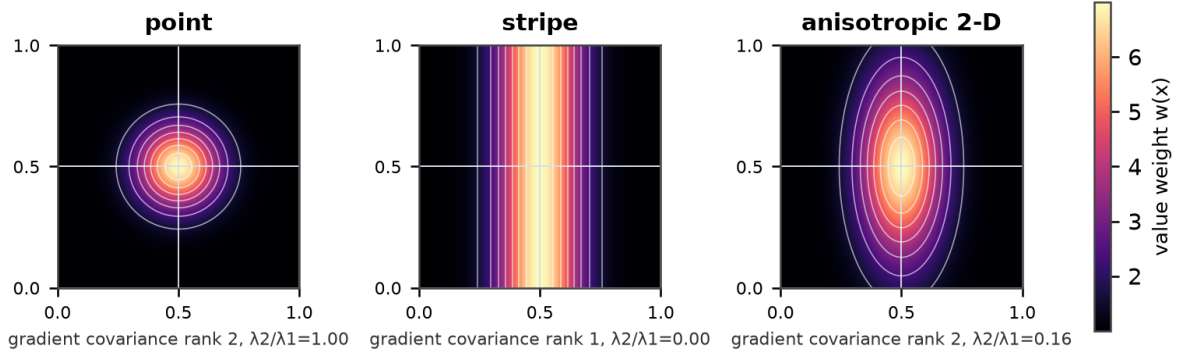


Figure 1. Registered value-weight fields $w(x)$. The stripe condition varies along one coordinate. The anisotropic 2-D condition has nonzero gradients along both arena axes; the plotted gradient-covariance diagnostic is rank 2 with $\lambda_2/\lambda_1=0.16$. Thus the failed 1/2 prediction is not explained by accidentally using a one-dimensional field.

The project-preregistered gate was frozen before the large Modal sweep. The primary estimand is the slope α from ordinary least squares regression of $\log \sqrt{\det g(x)}$ on $\log w(x)$, over spatial bins with finite positive area density. The primary comparison is at amplitude $A=6$, with A in $\{3, 6, 12\}$ used to check stability and amplitude scaling.

Confirmation of the 2-D law required an anisotropic 2-D mean closer to 1/2 than 1/3, a bootstrap standard error at or below 0.02, a stripe result near 1/3, and a nonzero aniso2d-vs-stripe gap. Failure of aniso2d to separate from stripe was preregistered as rejection of the $d=2$ gate; the effective-dimension interpretation was the planned diagnostic for such a failure.

4. Experiment

We trained 576 capacity-bottleneck path-integration RNNs on Modal H100 workers: three value-field geometries (point, stripe, anisotropic 2-D), three amplitudes (3, 6, 12), and 64 seeds per cell. Each network used $N_g=256$ hidden units, $N_p=256$ place cells, $T=20$ rollout steps, batch size 128, 8000 optimization steps, Adam with learning rate $1e-3$ and weight decay $1e-4$, unit-sphere hidden-state normalization at every recurrent step, and Gaussian channel noise with standard deviation 0.15 before decoding. Training weights were mean-normalized within each batch, so the intervention changes the spatial allocation of loss rather than the global loss scale.

Each condition-level α is the mean of 64 seed-level OLS slopes, not a pooled regression over all bins from all networks. Evaluation uses 20,480 sampled positions per model before 16×16 binning; finite-positive interior-bin fitting and empty-bin handling are reported in Appendix A.5.

Measured exponent landscape: all cells stay near the 1-D family

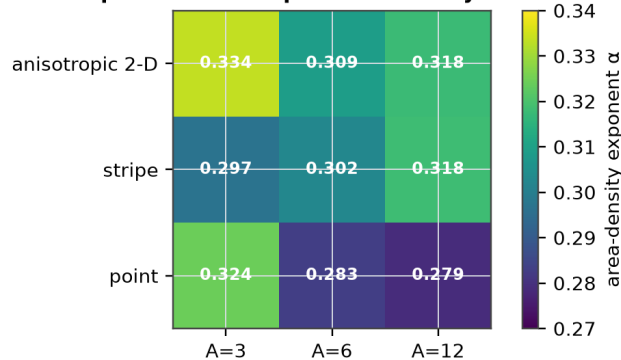


Figure 2. Full exponent landscape. Values are precise and stable, but they cluster around the 1-D family rather than the 2-D prediction.

5. Results

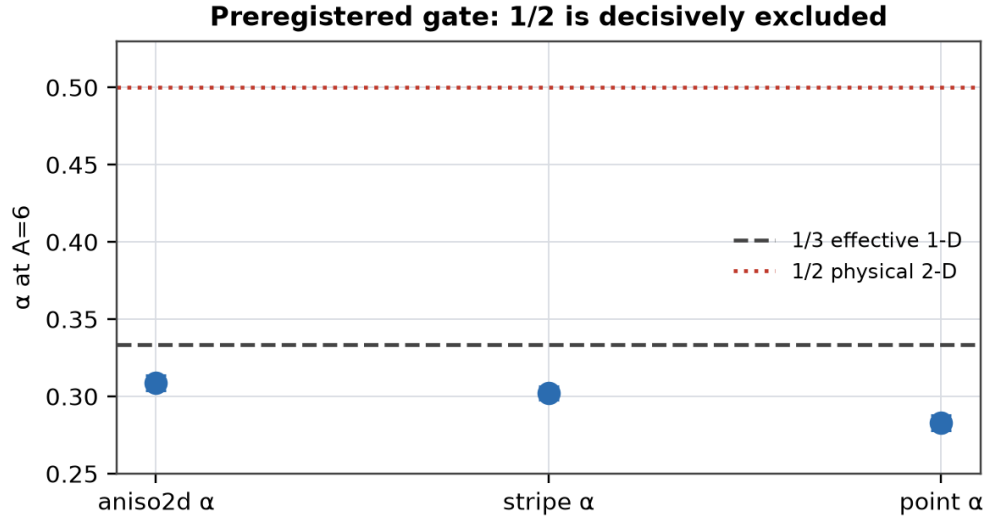


Figure 3. Primary preregistered gate. The anisotropic 2-D value field does not approach 1/2 and does not cleanly separate from stripe.

Geometry	A	α	95% CI	SE	d_eff	R ²
aniso2d	6	0.309	[0.304, 0.314]	0.0025	0.90	0.528
stripe	6	0.302	[0.298, 0.307]	0.0023	0.87	0.565
point	6	0.283	[0.278, 0.288]	0.0025	0.79	0.417

Table 1. Primary $A=6$ result. The 2-D exponent 1/2 is not within any interval.

The aniso2d-vs-stripe difference is $\Delta = +0.0065$ with bootstrap 95% CI $[-0.0003, +0.0132]$. That is not the expected separation between 1/3 and 1/2. The exponent landscape is therefore not a noisy confirmation of the 2-D prediction; it rejects the $d=2$, $\alpha=1/2$ prediction as the empirical description of this architecture.

Geometry	A=3 α	A=6 α	A=12 α	Reading
aniso2d	0.334 [0.329,0.338]	0.309 [0.304,0.314]	0.318 [0.313,0.322]	d_eff near 0.9-1.0
stripe	0.297 [0.291,0.302]	0.302 [0.298,0.307]	0.318 [0.314,0.322]	d_eff near 0.85-0.94
point	0.324 [0.318,0.330]	0.283 [0.278,0.288]	0.279 [0.275,0.283]	weaker radial allocation

Table 2. Full amplitude sweep. The exponent stays near the effective-1-D family.

Peak-resolution ratios increase monotonically with amplitude in all three geometries (point 1.319 to 1.400 to 1.515; stripe 1.289 to 1.348 to 1.428; anisotropic 2-D 1.300 to 1.369 to 1.443). Thus value weighting really does increase local resolution. The revised conclusion concerns the dimension and exponent governing that reallocation.

6. Estimator Sanity Check

Estimator sanity check at the experiment's 16x16 grid

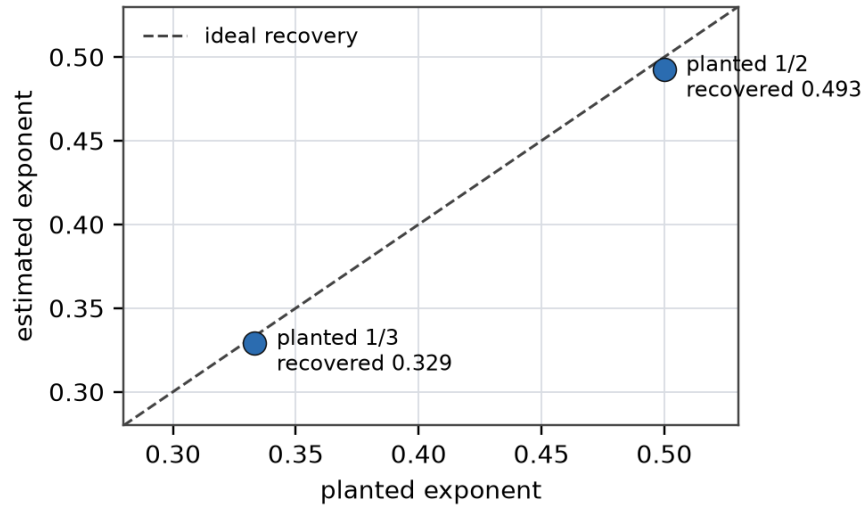


Figure 4. Synthetic estimator validation. A separable 2-D synthetic embedding with known area density $\rho(x)$ proportional to $w(x)^\alpha$ was evaluated with the same 16x16 central finite-difference area-density estimator and log-log slope fitter used on trained models.

Synthetic field	planted α	estimated α	R^2	bins
planted 1-D family	0.333	0.329	1.0000	196
planted 2-D law	0.500	0.493	1.0000	196

Table 3. The estimator does not collapse a planted 1/2 law to the observed 0.30 range.

This check is not evidence that the trained RNN should realize the 2-D optimum. It only addresses the simplest methodological objection: the area-density estimator and slope fitter, at the same 16x16 grid resolution, can recover planted 1/2 behavior and do not mechanically force slopes toward 1/3.

7. Interpretation

The exponent is not measuring the physical dimension of the arena. It is measuring the effective allocation dimension of the learned code. In this harness, d_{eff} is near one even when the externally specified value field varies along both physical dimensions.

The result changes the target for future theory. The missing ingredient is not precision: the primary standard errors are below 0.003. The missing ingredient is an architectural account of why recurrent grid-like codes spend capacity through a narrow degree of freedom. Possible explanations include radial-gradient allocation, modular grid periodicity, the unit-sphere bottleneck, and decoder geometry.

8. Limitations and NeurIPS-Scale Extensions

This standalone note is intentionally narrow. It establishes the preregistered negative gate and gives enough methodological detail to be auditable, but it does not yet claim universality across architectures or metric estimators. A full conference version should add capacity sweeps over N_g , noise, and normalization; architecture sweeps beyond this RNN harness; alternative metric estimators such as neighbor stretch, Frobenius norm, area density, and Fisher information; an ablation of unit-sphere normalization; and a decoder geometry ablation separating code geometry from readout geometry.

Appendix A. Methods

A.1 Environment and Trajectories

The arena is the unit square $[0,1]^2$ with reflecting boundaries. Each batch starts at a uniform random position in $[0.1,0.9]^2$. Heading is initialized uniformly on $[0,2\pi]$, perturbed at each step by Gaussian noise with standard deviation 0.4 radians, and moved at nominal speed 0.06 before reflection. Reflection flips out-of-bounds coordinates back into the arena and adds π to the heading for reflected particles. Rollout length is $T=20$.

A.2 Place-Cell Targets and Training Loss

$N_p=256$ place cells form a 16×16 grid of centers over the arena. The target place code is a softmax of negative squared distance to centers with sigma 0.09. The model receives the initial place code and velocity sequence, and predicts a place-code distribution at each step. The per-sample loss is $\text{KL}(\text{target} \parallel \text{predicted})$, multiplied by the mean-normalized value weight $w(x)$, and then averaged over batch and time.

A.3 Value-Weight Fields

All conditions use $w(x)=1+A \exp(-q(x))$ before mean normalization inside each training batch. For point, q is squared Euclidean distance from $(0.5,0.5)$ divided by $2\sigma^2$. For stripe, q uses only the x -coordinate distance. For anisotropic 2-D, q is the sum of an x term with width σ and a y term with width 2.2σ . The sweep used $\sigma=0.12$ and amplitudes A in $\{3,6,12\}$.

A.4 Model and Capacity Bottleneck

The model is a single RNNCell with ReLU recurrence. A linear encoder maps the initial place code into $N_g=256$ hidden units, the RNNCell integrates 2-D velocity inputs, and a linear decoder maps hidden states back to N_p place logits. After every recurrent update the hidden state is divided by its Euclidean norm plus $1e-6$, placing it on a unit sphere. During training, Gaussian channel noise with standard deviation 0.15 is added to hidden states before decoding. This unit-sphere plus finite-SNR channel is the finite-capacity bottleneck.

A.5 Area-Density Estimator

After training, each model is evaluated without channel noise on 1024 fresh trajectories. Hidden states are binned by position into the 16×16 place grid and averaged per bin, with empty bins filled by the mean occupied hidden state. For interior bins, central differences estimate the two positional derivatives du and dv from neighboring grid bins over $2\Delta x$. The estimated area density is $\sqrt{((du \cdot du)(dv \cdot dv) - (du \cdot dv)^2)}$, exactly the square root of the determinant of the local pullback metric.

At this 20,480-position evaluation scale, the 64 sweep seeds occupy all 256 spatial bins under the trajectory generator; mean occupied bins are 256/256 and interior occupancy is 100% (196/196 interior bins).

A.6 Slope Fitting and Bootstrap

For each trained network, α is the ordinary least squares slope in log area density versus log $w(x)$, restricted to finite positive interior bins. Each condition summary is the mean of 64 seed-level slopes. Confidence intervals and standard errors use 2000 bootstrap resamples of the seed-level rows within condition. The reported d_{eff} is computed from the condition mean using $d_{\text{eff}}=2\alpha/(1-\alpha)$.

A.7 Provenance

The gate was preregistered in the frozen 2026-07-02 addendum to the grid-cell weakness preregistration. The committed runner is `modal_reward_deformation_sweep.py`, and the

committed result report is reward_deformation_sweep_2026_07_02.md. Raw JSON artifacts are gitignored; the report records the manifest, summary statistics, and interpretation.

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Paper 3

Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization

Weakness, not generic compression alone, in shortcut-compatible learning problems

Included as part of the unified finite-representations research portfolio.

Symmetry-Compatible Hypothesis Volume Predicts Out-of-Distribution Generalization

Jawaun Brown

Weakness, not generic compression alone, in shortcut-compatible learning problems
Research-Derived Experiments · preprint compiled from the project repository

Abstract

We study finite-domain model selection where training data admits both a local shortcut and a globally symmetry-compatible rule. Loss and in-distribution validation cannot distinguish the completions, and generic simplicity or flatness proxies can prefer the shortcut. We define **weakness** as the number of transformations under which a candidate function remains equivariant up to an output action. On cyclic and dihedral symbolic families (500 trials each), weakness recovers the invariant rule in 100% of trials (Wilson 95% lower bound 0.992) while the tested train-loss, validation, description-length, MDL-style, compression, and flatness proxies recover it in 0% (upper bound 0.008). A **data-inferred** selector, built from circular-domain transformation enumeration plus training-pair consistency, matches the oracle without using the hidden offset or reflection parameter. Across 256, 1024, and 4096 trained MLPs, learned-function weakness under the true group is the strongest correlate of OOD accuracy ($r = +0.817, +0.813, +0.8085$; 4096 CI $+0.798$ to $+0.819$), and an augmentation-fixed-effect check remains positive (residual $r = +0.488$). Vision gives the same ordering (Z_8 rotation, $r = +0.672$). Parity is a clean negative case, and S_n is a large-group boundary case: oracle weakness helps, but data-inferred group discovery degrades. Pythia-70M is reported only in an appendix as latent geometry, not behavioral evidence.

1. Introduction

Modern pipelines are often underspecified: many predictors have equivalent training or in-distribution validation performance but different deployment behavior. Shortcut learning is one familiar failure mode. Here we make the ambiguity explicit: the training data is compatible with both a local shortcut and a globally transportable rule. If the deployment distribution is generated by transformations, the relevant question is not only which hypothesis is shortest, but which hypothesis remains valid under those transformations. This paper tests Bennett-style *weakness* — compatible transformation volume — as a finite, measurable selection rule.

2. Weakness

Let $f : X \rightarrow X$ be a candidate function on a finite domain of size n , and let G act on X . We call f *compatible* with $g \in G$ if there exists $h \in G$ with $f(g \cdot x) = h \cdot f(x)$ for all x (with-action equivariance; this generalizes strict equivariance $h = g$). Weakness is the count of compatible group elements:

$$W_G(f) = | \{ g \in G : \exists h \in G, \forall x, f(g \cdot x) = h \cdot f(x) \} |.$$

$W_G(f) = |G|$ means f is fully G -equivariant; $W_G(f) = 1$ means only the identity commutes. The *weakness selector* returns $\operatorname{argmax}_i W_{\hat{G}}(f_i)$ over a candidate pool, ties broken by the same compression proxy used by the baselines. Selection principle: when training constrains only part of a transformation orbit, train-consistent hypotheses compatible with more deployment-generating transformations should cover more of the missing orbit.

Tiny proposition: if deployment inputs are completed from observed examples by a subset S of G , then every compatible g in S transports observed labels along its orbit without contradiction.

More compatible transformations therefore cover more of the missing orbit, provided the candidate group is aligned with the deployment shift.

3. Symbolic separation

We build four task families, each with a known transformation group and a training distribution that admits both a train-perfect local shortcut and the true invariant: *cyclic_prefix_shift* (Z_n), *dihedral_reflection* (D_n), *parity_coset* (Z_2), and *color_permutation* (S_n). We compare eleven selectors over 500 trials per family with Wilson 95% intervals. Candidate pools are explicit: cyclic has $K=n+1$ (8/12/14 here), dihedral $K=3$, parity $K=4$, and color-permutation $K=3.6$ in the reported seeded run.

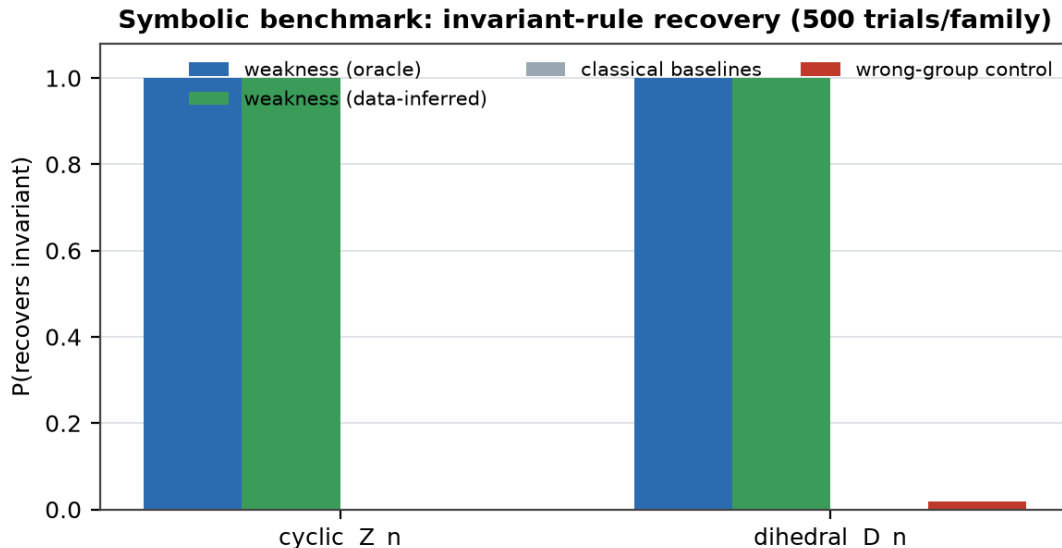


Figure 1. On cyclic and dihedral families, only weakness-based selectors recover the invariant rule (1.000; Wilson LB 0.992); training loss, simplicity, MDL, compression, flatness, and validation all recover it 0.000 (UB 0.008). The data-inferred selector uses circular-domain enumeration plus training-pair consistency, not the hidden offset/reflection, and matches the oracle. The wrong-group control is near zero.

Family	weakness (oracle)	data-inferred	classical baselines	wrong-group
cyclic Z_n	1.000	1.000	0.000	0.002
dihedral D_n	1.000	1.000	0.000	0.018
color S_n (partial)	0.824	0.138	0.000	0.000
parity Z_2 (negative)	0.000	0.000	0.000	0.038

Table 1. Invariant-recovery rate by selector and family. Cyclic and dihedral show a perfect, non-overlapping separation; S_n is a partial win; parity is a clean negative ($|G|=2$ too small to disambiguate).

4. Neural weakness predicts OOD accuracy

We train 256 small MLPs, replicate at 1024 models, and rescale to 4096 models via Modal on cyclic tasks with n in $\{7,11,13\}$, train window w in $\{2,3,4\}$, diverse depth, width, initialization, optimizer, learning rate, weight decay, and augmentation. From each model we extract the argmax function table and compute true-group weakness together with classical predictors, correlating each against held-out OOD accuracy.

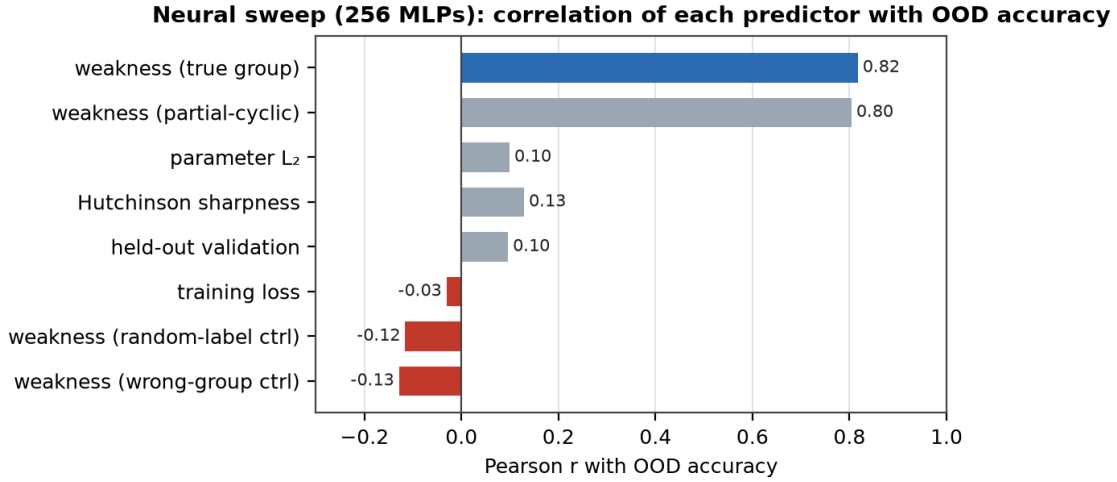


Figure 2. Across 256 MLPs, weakness under the true group is the single strongest predictor of OOD accuracy (Pearson $r = +0.817$ locally, $+0.813$ at 1024 models, $+0.8085$ at 4096 models). Training loss, validation, parameter L_2 , and sharpness are weaker; wrong-group and random-label controls are correctly negative.

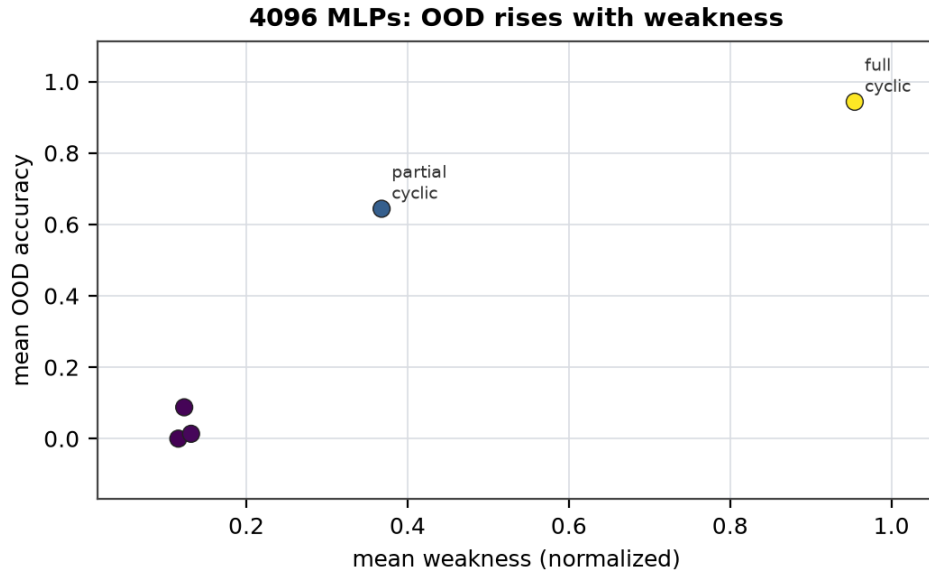


Figure 3. The per-augmentation gradient is monotone in weakness: full-cyclic (orbit completion) reaches 94% OOD and 0.95 weakness, while none / wrong-reflection collapse to $\leq 1.4\%$ OOD and ≤ 0.13 weakness. Augmentations that approximately respect the symmetry raise weakness and OOD together; after augmentation fixed effects, residual r remains $+0.488$.

5. Scaling: vision

On a synthetic Z₈ rotated-stroke task (eight classes, 16×16 , three of eight angles shown in training; a re-implementation of Perin and Deny's partial-orbit setup), weakness under the rotation group is again the dominant predictor across 96 models ($r = +0.67$), beating every classical predictor by $\geq 2\times$, with the pixel-permutation wrong-group control correctly anti-correlated ($r = -0.34$).

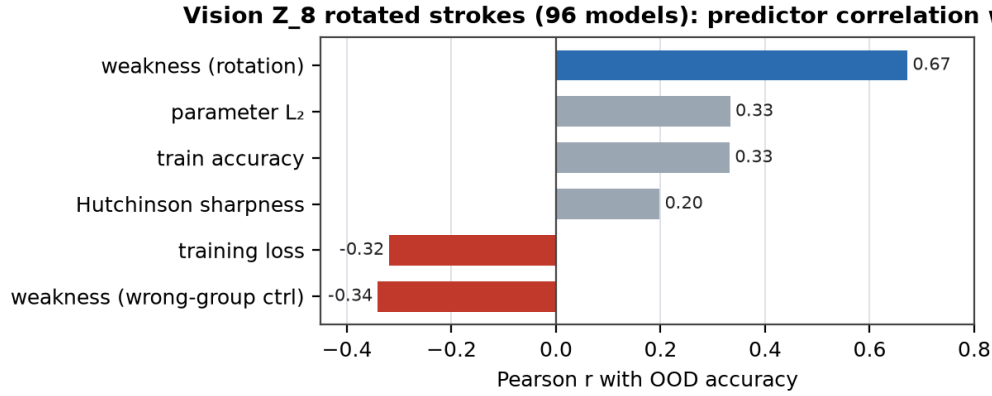


Figure 4. Vision (Z₈ rotation): weakness leads; the wrong-group (pixel-shuffle) control is anti-correlated, as expected of a model that has actually learned the rotation symmetry.

The language probe is moved to Appendix C and treated as exploratory latent geometry only; it is not used as behavioral OOD evidence.

6. Operating regime and limitations

Weakness ceases to discriminate when the candidate group is too small (parity, $|G|=2$) or too large/uninformative (full symmetric group, where wrong involutions have comparable centralizer-orbit sizes). This is a precise statement of when symmetry-volume is load-bearing, not a defect. Present limitations: small finite domains ($n \leq 13$ symbolic, 16×16 vision); intentionally simple MDL/compression/completion-volume and sharpness proxies; the language latent→behavior chain unconfirmed; transformation discovery is still enumerative. The natural next steps are stronger compression/PAC-Bayes baselines, learned transformation generators, and training-time weakness regularization.

7. Discussion

The discriminating quantity between local shortcut and globally invariant rule — on families where both are train-perfect — is symmetry-compatible-hypothesis volume: a measurable, intervention-friendly, reparameterization-invariant quantity, not a parameter-space artifact. It operationalizes Bennett's weakness on neural function tables and answers the practical question of which heuristic to trust when training loss is tied. The safe thesis is not that compression is wrong; it is that, in these shortcut-compatible symmetry tasks, the relevant compression is compatibility with the transformations that generate the missing cases.

Appendix A. Methods and statistics

Symbolic families are exact finite tasks. Cyclic uses $X=Z_n$, n in $\{7,11,13\}$, truth $f_b(x)=x+b \bmod n$, a prefix train window, and a suffix OOD set. Dihedral uses $X=Z_n$, n in $\{7,9,11,13\}$, truth $f_b(x)=b-x \bmod n$, and D_n rotations/reflections. Parity uses an even domain and $f(x)=x \text{ xor } 1$ with one parity coset held out. Color-permutation uses S_n , n in $\{4,5,6\}$, a sampled non-identity permutation, sparse training inputs, and unobserved inputs as OOD.

Selector	Score	Tie-break	Group access
train_loss	train accuracy	shorter form	none
validation	leave-one-observed-pair accuracy	shorter form	none
simplicity	shorter form_length	random exact tie	none
compression	form_length + 20*train_errors	random exact tie	none

mdl_program	2 [^] -form_length	shorter form	none
flatness_proxy	completion-volume proxy, not Hessian flatness	shorter form	none
weakness_oracle	W_G(f)	compression	true task group
weakness_wrong_group	W under random perms	compression	wrong control
weakness_noisy_group	W under noisy subset	compression	noisy true group
weakness_data_inferred	W under inferred G_hat	compression	train pairs only
random	uniform train-perfect candidate	-	none

Appendix Table A1. Exact selector definitions. Classical baselines are the first six rows; validation is in-distribution only.

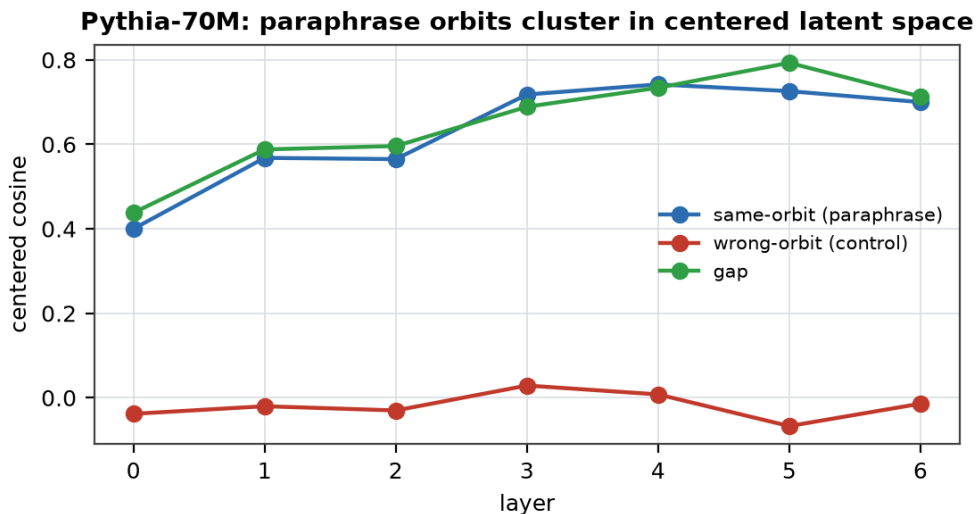
Data-inferred group discovery enumerates a family-specific domain prior (Z_n shifts for cyclic, D_n rotations/reflections for dihedral, parity swap for parity, and a small cyclic heuristic for S_n), then keeps every input-side transformation g for which some output-side transformation h is non-contradictory on observed training pairs. It uses training inputs, training outputs, and the domain prior only: no test labels, hidden offset, hidden reflection, sampled permutation, or invariant-candidate identity.

Statistic	Value
4096 true-group Pearson r	+0.8085 (95% CI +0.7976 to +0.8189)
4096 partial-cyclic Pearson r	+0.7940 (95% CI +0.7824 to +0.8051)
4096 validation Pearson r	+0.0924 (95% CI +0.0620 to +0.1227)
4096 parameter L2 Pearson r	+0.2533 (95% CI +0.2244 to +0.2818)
4096 fixed-effect residual r	+0.4883 (95% CI +0.4647 to +0.5113)
Fixed-effect beta	+0.3652 (95% CI +0.3452 to +0.3853)

Appendix Table A2. Fisher-z intervals and augmentation fixed-effect check. The headline effect is partly augmentation-mediated, but a within-condition learned-function signal remains.

Appendix C. Exploratory language check

Pythia-70M hidden states for 24 concepts x 3 paraphrases show centered same-orbit cosine gaps of +0.44 to +0.79 after All-but-the-Top centering, but per-concept next-token behavioral consistency is not predicted at this scale (Pearson range -0.35 to -0.13, $N=24$). This supports only a latent-geometry statement: paraphrase orbits cluster in centered representation space.



Appendix Figure C1. Pythia-70M paraphrase orbits cluster in centered latent space (gap peaks at +0.79, layer 5). This is a latent-geometry result only.

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Paper 4

Translation Augmentation Produces Toroidal Codes and Larger-Arena Generalization in Path-Integration RNNs

Negative mediation result for weakness as the governing scalar

Included as part of the unified finite-representations research portfolio.

Translation Augmentation Produces Toroidal Codes and Larger-Arena Generalization in Path-Integration RNNs

Jawaun Brown

Research-Derived Experiments · empirical note and negative mediation result

Abstract

This note tests a pre-registered strong claim about path-integration RNNs: **weakness**, a held-out linear equivariance score for wrapped translations, should govern whether a learned population code forms a toroidal manifold, and toroidal topology should mediate larger-arena generalization. The result is a useful negative mediation result. In a 320-network Modal H100 sweep, full translation augmentation reliably induces toroidal codes (G1 pass; torus match 0.734), preserves decoding when the arena doubles (OOD 0.949 versus 0.484 for no augmentation), and produces the causal condition contrast specified before the sweep (G6 pass). Weakness also tracks spectral concentration (G5 pass; $\rho = +0.635$). But the central triangle fails: weakness only weakly predicts toroidal score (G2 fail; $\rho = +0.197$), does not beat final loss on larger-arena OOD by the pre-registered 2x margin (G3 fail; weakness $\rho = +0.617$, loss $\rho = +0.652$), and topology does not mediate the weakness-OOD relation (G4 fail). The surviving claim is therefore bounded: in this artificial RNN harness, translation-structured training causally produces toroidal, OOD-generalizing codes, while weakness measures one spectral aspect of the learned translation structure rather than serving as the governing scalar of torus formation.

1. Framing

Biological grid-cell population activity has been shown, using topological data analysis, to lie on a toroidal manifold, and path-integration RNNs can learn grid-like spatial codes. This experiment asks a bridge question: when toroidal codes and larger-arena generalization appear in a learned path integrator, are they governed by the same substrate-general weakness quantity that predicted OOD behavior in the symbolic and vision experiments, or are they more directly caused by the translation-structured training condition itself?

The strong version predicted a triangle: high weakness, low Fourier participation ratio (few aligned frequency modes), clean toroidal topology, and larger-arena OOD generalization should be different measurements of one event. The sweep was designed to make that strong triangle falsifiable. We therefore present the result as an empirical note and negative mediation result, not as a broad confirmation of the weakness program.

2. Pre-Registered Test

The gates were specified in the project repository before the Modal sweep (`papers/grid\_cell\_weakness/preregistration.md`, frozen 2026-06-28). The registration was project-level rather than an external registry filing. It asked whether a single translation-compatibility scalar predicts both population topology and OOD accuracy, and whether topology carries the weakness-to-OOD relation.

Gate	Pre-registered criterion	Modal result	Verdict
G1 manifold	full translation torus match ≥ 0.60	0.734	pass
G2 weakness-topology	$\rho \geq 0.5$ and $\geq 2\times$ best baseline	$\rho +0.197$; loss $+0.431$	fail
G3 weakness-OOD	$\rho \geq 0.5$ and $\geq 2\times$ best baseline	$\rho +0.617$; loss $+0.652$	fail
G4 mediation	partial ρ drops by $\geq 50\%$	partial $\rho +0.623$; no drop	fail
G5 spectrum	$\rho(\text{weakness, -Fourier PR}) \geq 0.5$	$+0.635$	pass
G6 causal	full translation beats none/random shift	topology and OOD both lift	pass
wrong-group null	wrong-group predictor null or negative	tie-aware $\rho 0.000$	pass

Table 1. Locked gates and outcomes. The decisive failure is not auxiliary: G2-G4 reject the registered mediation triangle.

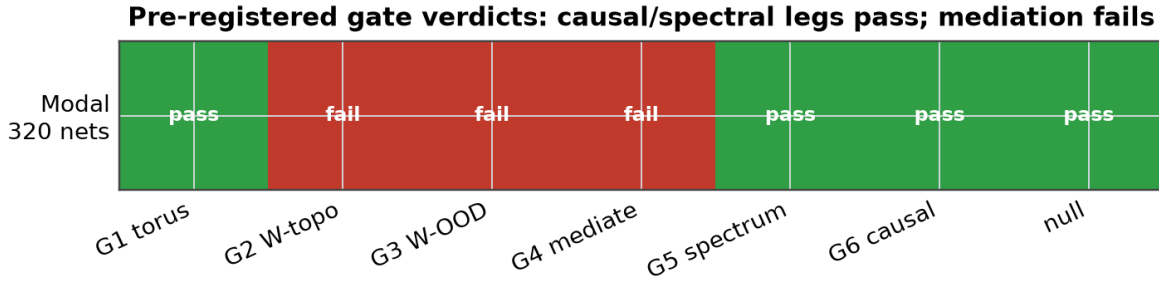


Figure 1. Gate verdicts from the pre-registered Modal sweep. The causal and spectral legs pass; the topology-governing and mediation claims fail.

3. Methods

We trained 320 velocity-driven path-integration RNNs on Modal H100 workers: five training conditions, two recurrent architectures, and 32 seeds. Each network predicts a place-cell code from velocity sequences. Conditions vary the augmentation regime: full translations, partial translations, random shifts, no augmentation, and a wrong-group control. For each trained code we measure weakness, persistent-homology toroidal score, Fourier participation ratio, and decoding accuracy in held-out arenas of scale 1.0, 1.25, 1.5, and 2.0. The largest arena is the pre-registered OOD metric.

Weakness. For a binned hidden-state population code $H(x)$ and a held-out wrapped translation τ , fit one least-squares linear operator A_τ on half the spatial bins and evaluate it on held-out bins: $W = E_\tau R^2[H(x + \tau), A_\tau H(x)]$. Translations are wrapped modulo the 16×16 bin grid, one operator is fit per translation, R^2 is computed over hidden units and held-out positions, and the reported weakness is the mean clipped R^2 across the registered shifts. The metric is computed on hidden activity, not on the decoded position or place-cell target.

Statistics. Unless otherwise stated, ρ denotes Spearman rank correlation with average-rank tie handling. G2 and G3 report signed ρ values, while the registered $2\times$ baseline comparisons use absolute ρ for classical predictors. The loss baseline is raw final training loss, not negative loss; its positive ρ with OOD reflects the fact that the translation-augmented condition is harder to fit but generalizes better.

Topology. Hidden states are sampled from fresh trajectories, averaged into a 16×16 spatial grid, and treated as a point cloud in hidden-state space. Empty spatial bins, when present, are filled with the global mean so the grid is complete; the Modal cells report coverage. Persistent homology is computed with a Vietoris-Rips complex (Gudhi backend, max simplex dimension 3, edge cap at the 45th percentile of pairwise distances), yielding H_0 , H_1 , and H_2 persistence intervals. A torus should have Betti signature (1,2,1): one component, two loops, and one void.

The continuous toroidal score combines the second H1 bar above the third-bar noise floor with the strongest H2 void; ``betti_match_torus`` requires two estimated H1 loops and a nontrivial H2 bar. Torus match is the fraction of networks in a condition satisfying this Boolean criterion.

4. Results

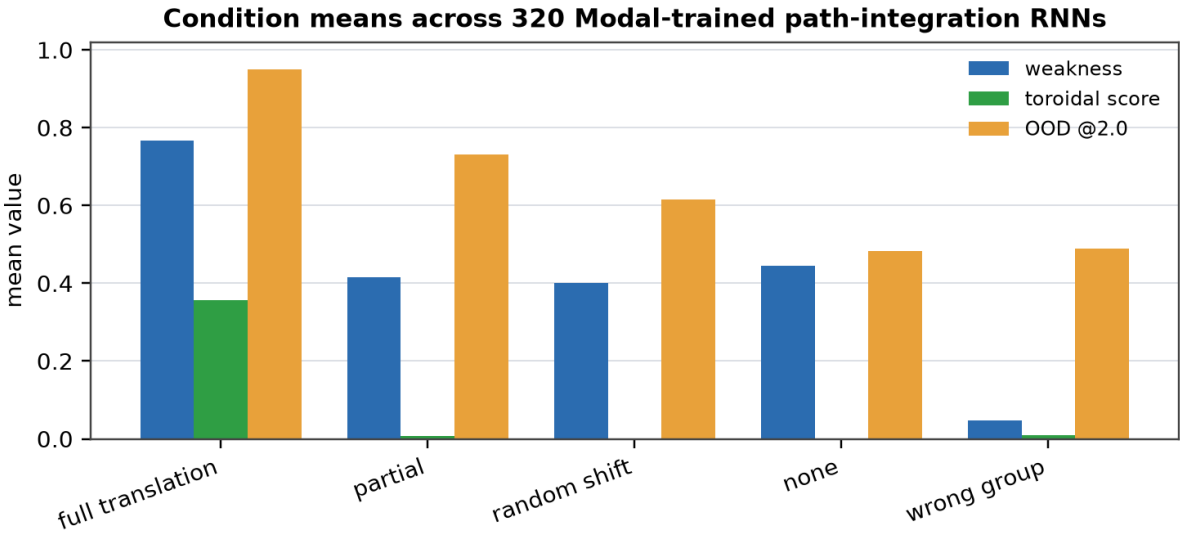


Figure 2. Condition means. Full translation is the only condition that reliably forms a torus and preserves OOD decoding at arena scale 2.0.

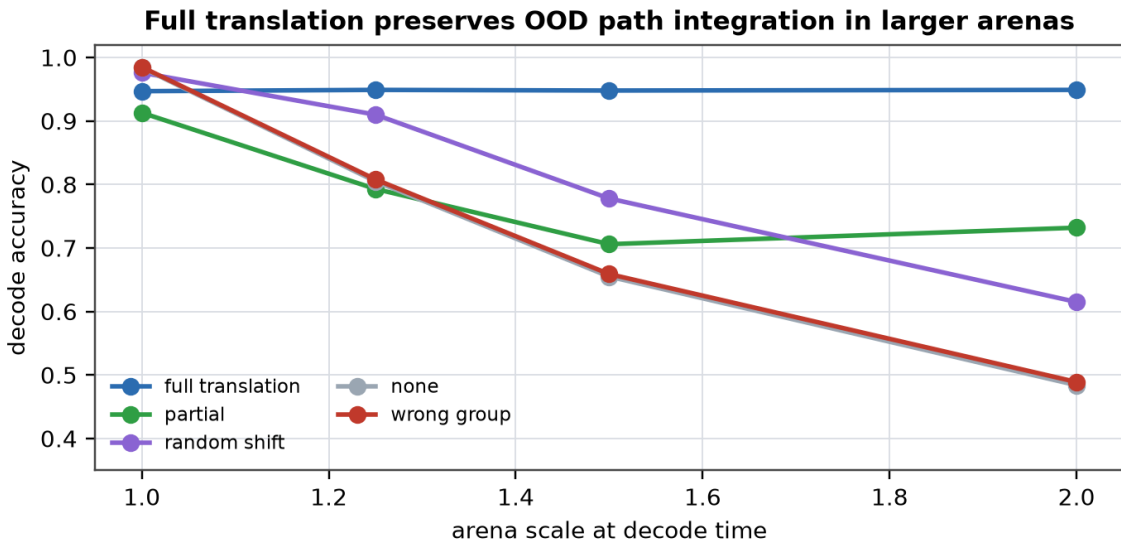


Figure 3. Larger-arena OOD curves. Full translation remains stable as the arena doubles; controls degrade sharply.

Condition	n	weakness	toroidal	torus match	Wilson 95%	OOD @2.0
full translation	64	0.768	0.357	0.734	[0.61, 0.83]	0.949
partial translation	64	0.416	0.007	0.000	[0.00, 0.06]	0.732
random shift	64	0.400	0.000	0.000	[0.00, 0.06]	0.615
none	64	0.446	0.000	0.000	[0.00, 0.06]	0.484
wrong group	64	0.048	0.009	0.000	[0.00, 0.06]	0.489

Table 2. Condition means. Wilson intervals apply only to the torus-match fraction. Full translation is the positive intervention; random shift and wrong group are controls. Weakness is not monotone with toroidal score outside the full-translation condition.

The recovered raw-cell export also supports seed-level uncertainty for the continuous metrics. Appendix B reports percentile bootstrap intervals for weakness, toroidal score, Fourier participation ratio, and each OOD arena.

The positive result is clean at the condition level. Full translation augmentation is the only condition that reliably produces a torus, and it preserves decoding accuracy as the arena doubles. Partial translation and random shifts lift OOD relative to no augmentation, but they do not produce the torus match. The wrong-group control has near-zero weakness under the registered translation metric and does not lift topology or OOD.

The negative result is equally important. Across individual networks, weakness does not explain toroidal score strongly enough to pass the registered gate, does not outperform final loss on OOD by the required margin, and leaves the weakness-ODD association intact after controlling for topology. These are not wording problems; they reject the strongest version of the theory tested here.

5. Interpretation

The result should not be polished into a full confirmation. It is better than that: it marks the boundary of the weakness program on a task where topology is genuinely load-bearing. Translation augmentation is the experimentally manipulated variable that produces toroidal topology and larger-arena generalization in this harness. Weakness is still useful as a spectral and OOD-associated measurement, but it is not the scalar that governs torus formation here.

A plausible reading is that the augmentation condition itself is the strongest causal variable: it directly supplies the group structure needed for both topology and OOD. Weakness measures part of that event, especially its spectral footprint, but it is not sufficiently specific to explain topology variance after training loss and condition effects are present. The failed gates are not auxiliary failures; they reject the strongest claim. The surviving claim is not that weakness governs toroidal topology, but that translation-structured training produces toroidal, OOD-generalizing codes while weakness tracks one spectral aspect of learned translation structure.

6. Scope and Limitations

This remains an artificial RNN path-integration task, not a biological replication. The paper should not claim to explain biological grid cells. Its claim is narrower: a learned path-integration model, inspired by the grid-cell/topology literature, responds to a translation-augmentation intervention by forming toroidal hidden-state structure and generalizing to larger arenas. The topology metric uses binned hidden activity and Vietoris-Rips persistent homology; both can be sensitive to sampling density, binning, edge thresholds, and seeds. The wrong-group null in the first Modal summary was miscomputed by a rank helper that did not average ties; the corrected tie-aware Spearman is reported here and in the result report. The raw cell measurements did not change.

Future tests should separate condition effects from within-condition variation by sampling a larger family of already-toroidal codes, perturbing trained codes rather than only training regimes, and checking topology robustness under multiple bin counts, point sampling strategies, and persistent-homology backends. A biological extension would be a new confirmatory study on public grid-cell recordings, not an interpretation of the RNN result itself.

Appendix A. Implementation Details

Component	Vanilla RNN cell	GRU cell
recurrent update	torch RNNCell, ReLU	torch GRUCell + ReLU hidden
hidden size	128	128
input at t	2-D velocity	2-D velocity
initial state	linear encoder of initial place code	same
output	100 place-cell logits	same
loss	KL divergence to target place-cell code + 1e-3 activity penalty	same
optimizer	Adam, lr 1e-3, weight decay 1e-4	same
training	4000 steps, batch 200, trajectory length 20	same

Table A1. Architecture and training hyperparameters used in every Modal cell.

Path-integration task. Each training trajectory starts with x_0 sampled uniformly from the interior $[0.1B, 0.9B]^2$ of a square box of side B . Heading is initialized uniformly and perturbed each step by $\text{Normal}(0, 0.4)$; velocity is $0.06(\cos \theta, \sin \theta)$. Positions update as $x(t+1) = x(t) + v(t)$ with reflecting boundaries. Training uses $B=1.0$. Place targets are softmax-normalized Gaussian activations over a 10×10 unit-square place-cell grid with $\sigma=0.10$.

Condition	Exact intervention in the worker
none	no augmentation
full translation	sample offset $u \sim \text{Uniform}([0,1]^2)$; add u to positions and initial position modulo 1; velocities unchanged
partial translation	sample offset $u \sim \text{Uniform}([0,0.3]^2)$; add u modulo 1; velocities unchanged
random shift	sample offset $\epsilon \sim \text{Normal}(0,0.05)$; add to positions and initial position, clipped to the current box
wrong group	swap the two velocity coordinates while leaving target positions unchanged
null predictor	separate wrong-group metric: replace wrapped translations by a fixed random bin permutation in the weakness calculation

Table A2. Training conditions and the separate wrong-group null predictor.

OOD decoding. Evaluation generates fresh trajectories with B in $\{1.0, 1.25, 1.5, 2.0\}$. The model's place-cell argmax is counted correct when its center lies within one unit-square place-cell spacing of the target argmax. The primary OOD score is the largest scale, $B=2.0$. The same condition-specific preprocessing is used at decode time; therefore the result should be read as larger-trajectory/arena-scale OOD in this harness, not as an unbounded coordinate extrapolation claim.

Fourier participation ratio. Hidden activity is averaged into 16×16 spatial rate maps per unit. After subtracting each map's mean and dropping the DC Fourier bin, power is normalized over spatial frequencies and $PR = 1 / \sum_k p_k^2$ is averaged over units. Lower PR means fewer effective Fourier modes; G5 correlates weakness with -PR.

Topology and uncertainty status. The committed result report stores condition means and gate correlations; the recovered raw per-cell JSON has now been exported to committed CSVs and supports seed-level bootstrap intervals for scalar metrics. It does not store the hidden-state populations needed to reconstruct topology robustness over bin counts, Vietoris-Rips edge caps, empty-bin handling, or sampling density. The Modal runner now supports that robustness export for reruns, but the present PDF does not treat robustness as completed evidence.

Appendix B. Conference Evidence Exports

The raw 320-cell Modal JSON was recovered locally and exported with ``scripts/analyze_gridcell_conference_evidence.py`` into per-cell and aggregate CSVs under ``experiments/grid_cell_weakness/results``. Continuous intervals are percentile bootstrap 95% intervals from 5000 resamples within condition; torus-match intervals are Wilson 95% intervals for the Boolean ``betti_match_torus`` fraction.

Condition	weakness	toroidal	Fourier PR	OOD @2.0
full translation	0.768 [0.723, 0.808]	0.357 [0.317, 0.396]	4.472 [4.188, 4.773]	0.949 [0.946, 0.953]
partial translation	0.416 [0.363, 0.467]	0.007 [0.006, 0.009]	7.557 [7.094, 8.038]	0.732 [0.725, 0.738]
random shift	0.400 [0.368, 0.433]	0.000 [0.000, 0.000]	8.778 [8.200, 9.354]	0.615 [0.597, 0.628]
none	0.446 [0.409, 0.481]	0.000 [0.000, 0.000]	8.324 [7.791, 8.899]	0.484 [0.473, 0.495]
wrong group	0.048 [0.033, 0.064]	0.009 [0.007, 0.011]	14.634 [14.128, 15.180]	0.489 [0.479, 0.499]

Table B1. Scalar metric means with 95% intervals. Fourier PR is included because the spectral leg is the surviving weakness-aligned measurement.

Condition	1.0	1.25	1.5	2.0
full translation	0.947 [0.944, 0.949]	0.949 [0.946, 0.952]	0.948 [0.945, 0.951]	0.949 [0.946, 0.953]
partial translation	0.913 [0.909, 0.918]	0.793 [0.786, 0.800]	0.706 [0.697, 0.715]	0.732 [0.725, 0.738]
random shift	0.976 [0.958, 0.987]	0.910 [0.890, 0.923]	0.778 [0.756, 0.793]	0.615 [0.597, 0.628]
none	0.984 [0.980, 0.987]	0.805 [0.794, 0.815]	0.655 [0.645, 0.665]	0.484 [0.473, 0.495]
wrong group	0.985 [0.983, 0.987]	0.808 [0.797, 0.818]	0.659 [0.650, 0.669]	0.489 [0.479, 0.499]

Table B2. OOD decoding curves with bootstrap 95% intervals.

Within-toroidal subset. Among the 47 already-toroidal full-translation models, weakness does not explain additional OOD variation after torus formation: $\rho(\text{weakness}, \text{OOD}) = -0.198$ with bootstrap 95% CI [-0.518, 0.136]. Within this subset, weakness also anticorrelates with continuous toroidal score ($\rho = -0.335$, CI [-0.577, -0.063]) and with -Fourier PR ($\rho = -0.356$, CI [-0.585, -0.071]). No control condition has enough torus-matching models for the same analysis.

Topology robustness. The recovered scalar raw JSON cannot reconstruct robustness over bin counts, edge caps, empty-bin handling, or sampling density because it does not store hidden-state populations. The runner now emits ``topology_robustness`` rows when rerun with robustness enabled; until then, robustness remains the next required conference-review evidence item.

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